

Aradhya Agarwal

aradhyeagarwal@gmail.com

EDUCATION

Indian Institute of Technology Delhi

Delhi, India

M.Tech. in Computer Science and Engg. **GPA: 9.00/10.00**

Sep 2023 – Sep 2025

B.Tech. in Computer Science and Engg. **GPA: 8.26/10.00**

Sep 2020 – Sep 2024

WORK EXPERIENCE

Research Fellow at Microsoft Research India

July 2025 – ongoing

- Building safe agents capable of rejecting harmful queries or blocking tool calls which are found to be harmful as the tool calling chain unrolls
- Supervised by principal researcher Akshay Nambi, who leads the reasoning team at MSR India

PUBLICATIONS

Guarding Agentic Reasoning Models for Safe Multi-Step Tool Use

International Conference on Machine Learning (ICML) 2026

Aradhya Agarwal, Gurdit Siyan, Yash Pandya, Joykirat Singh, Akshay Nambi, Ahmed Awadallah

Step-by-Step Unmasking for Parameter-Efficient Fine-tuning of Large Language Models

Transactions of the Association for Computational Linguistics (TACL) 2024

Aradhya Agarwal*, Suhas K. R.*, Ayan Sengupta*, Tanmoy Chakraborty

ECoDepth: Effective Conditioning of Diffusion Models for Monocular Depth Estimation

Computer Vision and Pattern Recognition (CVPR) 2024

Suraj Patni*, **Aradhya Agarwal***, Chetan Arora

The Third Monocular Depth Estimation Challenge

Computer Vision and Pattern Recognition Workshop (CVPRW) 2024

Jamie Spencer, [...], **Aradhya Agarwal**, [...], James H. Elder

First Finish Search: Efficient Test-Time Scaling in Large Language Models

Aradhya Agarwal, Ayan Sengupta, Tanmoy Chakraborty

The Art of Scaling Test-Time Compute for Large Language Models

Aradhya Agarwal, Ayan Sengupta, Tanmoy Chakraborty

INTERNSHIPS

SWE at Google India

May 2024 – July 2024

- Worked from the Hyderabad office to enhance and automate a contract generation system, streamlining workflows and reducing manual effort
- Significantly reduced the SWE work effort in executing repetitive tasks from 5 working days to just 2

AWARDS & ACHIEVEMENTS

Won USD 10,000 in Kaggle's "Child Mind Institute – Problematic Internet Usage," placing second

December 2024

One of 50 students from across India selected to attend the Google DeepMind Research Symposium

January 2025

Invited to the OpenAI Codex Hackathon in Bangalore, India

April 2026

Invited to the xAI hackathon in SF, CA

December 2025

TALKS

The Third Monocular Depth Estimation Challenge: CVPR'24 Workshop

Jun 2024

- One of the winners of the MDEC challenge with the work featuring in the workshop paper
- Invited to deliver a talk at the workshop due to the novelty of the method

Thesis Talk at IIT Delhi

May 2025

RELEVANT COURSEWORK

Deep Learning, Natural Language Processing, Linear Algebra, Probability Theory and Stochastic Processes, Digital Logic, System Design, Data Structures, Computer Architecture, Programming Languages, Optimization Methods, Operating Systems, Computation Theory, Machine Learning, Networks, Discrete Mathematics, Artificial Intelligence, Algorithm Design, Logic for Computer Science, Numerical Methods